VoxPreference: a design review of the automated Preference Grammar tool

Demilade Bickersteth · Covenant University · 20 October 2026

Prof. C.U.C. Ugorji's Preference Grammar picks a preferred pronunciation from the forms speakers actually use, by ranking them on six tenets. Done by hand it takes a great many hours, so the team set out to automate it: our ICAEISD 2024 paper and slides, Prof's framework for the automation, the PRD, and the user flow in Figure 1. My final-year project was one block of that plan, the speech-to-text conversion.

This report reviews the whole planned tool, component by component: what the plan said, what made sense, what did not and why, and what we found and built instead. Then it gives the results. The tool now runs end to end, from recordings to the operability table, is live on the web, and is reliable today for voiceless and voiced <th> and the KIT vowel.

1. Preference Grammar, in Ugorji's terms

PG has three components. The input is every attested form of a sound, in its words, as speakers produce it, with no underlying forms. The parameters are the six tenets: International acceptance, Contrastiveness, Frequency, Disambiguity, Phonetic simplicity and Pedagogic convenience, ranked by default as I-Ac >> ContR >> FreQ >> DisaM >> Pho-S >> Ped-C. As in Optimality Theory the ranking is strict but violable, and a user may promote, demote or delete a parameter. The output is the preferred forms, the candidates that earn the fewest violations. The working is shown in an operability table. The published examples are *think*, where [θɪŋk] beats [tɪŋk], [sɪŋk] and [zɪŋk], and the GOAT vowel, where [o] wins. One outcome of PG is a Nigerian national model of the vowels.

So the automated tool has one job: take recordings to attested forms, attested forms to marks on six ranked tenets, and marks to the operability table and the preferred form. Any automation has to reproduce Ugorji's two worked examples.

2. What we found, in short

- **The central mistake was converting speech to text and then text to IPA.** That answers "what was said", but PG asks "how was it said". Because the output always looked plausible, this hid every other fault.
- **Clustering was asked to discover pronunciations** from features that mostly carry the speaker and the microphone.
- **The tenets had definitions but no evidence.** Without a rule for each mark, nothing could be ranked.
- **Success was measured by word error rate.** For PG what matters is how well the variant shares are estimated, checked against human labels.
- **The infrastructure was planned before the method worked.**

Hear the sound directly, give every tenet its evidence, and measure against human ears.

3. The planned tool

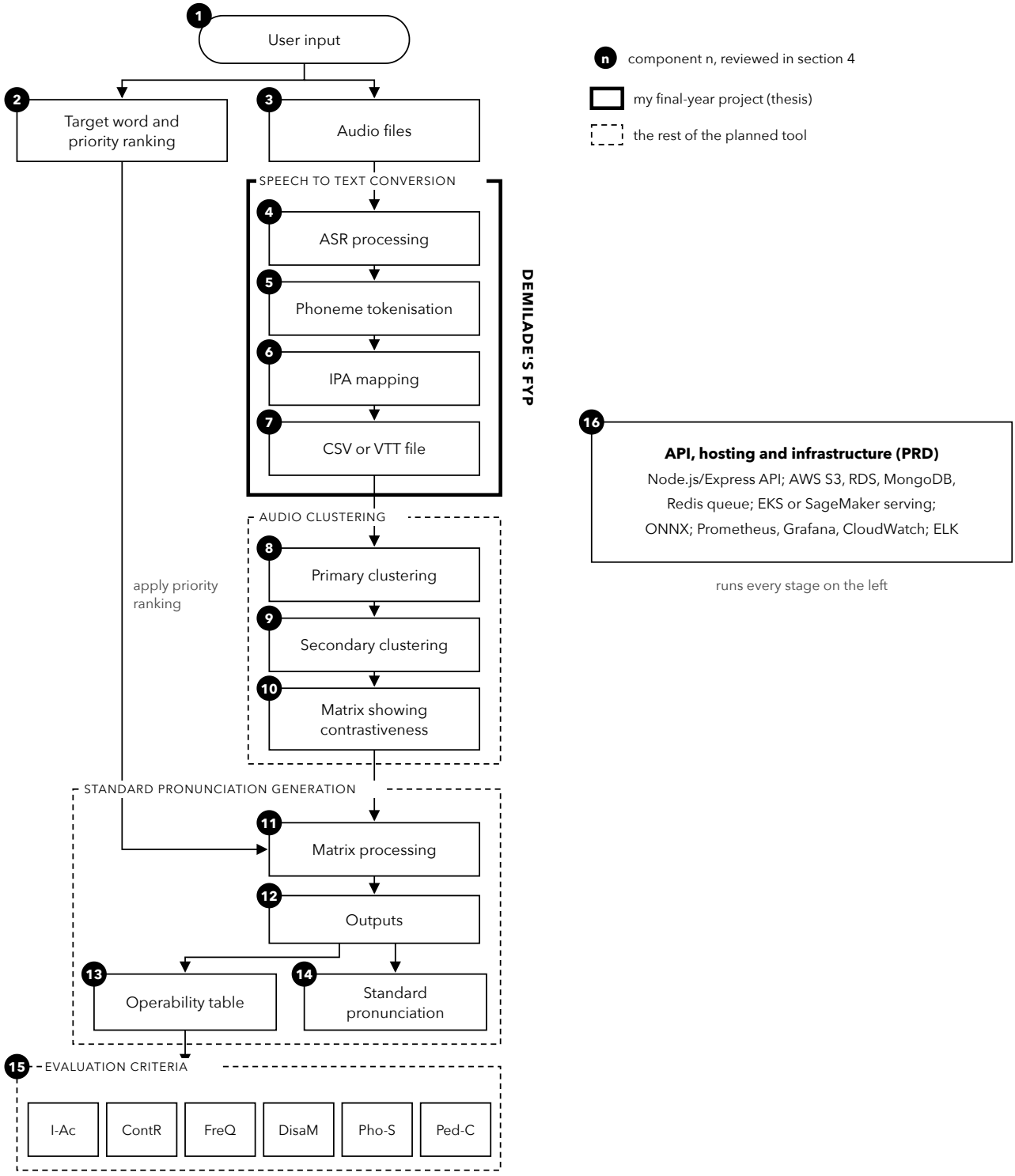


Figure 1. The planned automated PG tool: the team's user flow for the Preference Grammar system, redrawn, with the PRD's infrastructure beside it. The numbers match the components in section 4. The block in the thick outline, speech-to-text conversion, was my final-year project.

The aim was right. Most of what went wrong sat between the boxes: what each one had to hand to the next.

4. Component by component

Input

1 User input

THE PLAN	A web system where a user uploads speech and sets a tenet preference; the PRD adds user stories for researchers, educators, developers, policy makers and learners.
MADE SENSE	Putting the user's ranking at the centre. Ugorji's PG lets a user promote, demote or delete parameters, so the tool has to let them.
DID NOT, AND WHY	Five audiences and their features were planned before one analysis worked end to end.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	One page for one job: words, recordings and a ranking in; the operability table and the preferred form out. Everything else can wait for that to be trusted.

2 Target word and priority ranking

THE PLAN	The user gives a target word and a priority ranking; the ranking is applied at matrix processing.
MADE SENSE	PG's input is attested forms within lexical contexts, and the word is that context. The ranking belongs to the user.
DID NOT, AND WHY	One word gives few tokens. The plan did not say which sound in the word is under study, and a ranking as a plain list cannot pool two tenets or delete one, which PG allows.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	Several words that share a sound; the sound is chosen from a list (or inferred) and each is marked validated or not. The ranking editor reorders, deletes and pools tenets.

3 Audio files

THE PLAN	Accept large continuous speech files. Prof's framework reviews continuous speech separation for overlapping talk.
MADE SENSE	Real speech from many speakers, including long recordings, is what PG needs.
DID NOT, AND WHY	Every file was treated as one voice. Separation was reviewed in the framework but never designed, so a podcast panel would have counted as one speaker.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	Any common audio format, voice notes recorded in the browser, and podcasts. Speaker separation (ECAPA) credits each word to the person who said it (section 6).

Speech to text conversion: my final-year project

4 ASR processing

THE PLAN	PRD: a CNN-RNN trained on Colab with MFCC features. My thesis: wav2vec2 fine-tuned on Nigerian English (SLR70: 5.77 hours, 3,359 utterances, 31 speakers).
MADE SENSE	Starting from a pretrained wav2vec2 model and public Nigerian recordings. Training a CNN-RNN from scratch on about 6 hours of audio would not have worked.

DID NOT, AND WHY	The fine-tune never trained. The transcripts were lowercased against an uppercase vocabulary, so all 174,915 label tokens became `<unk>`. transformers 4.51.3 left `masked_spec_embed` uninitialised: every loss was NaN, logged as 0.000000, and the learning rate stayed at 0 for 3,780 steps. The published model is byte-for-byte `facebook/wav2vec2-base-960h`. The random 90/10 split put all 31 speakers in both halves and 67 of 336 validation sentences (20%) in training too. The stock model scores WER 36.7% there; the thesis reports 41.10% with the same 112 deletions, so it very likely measured the stock model. Underneath all of it, ASR answers which words were said, and PG asks how.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	ASR only finds the words. Whisper small searches unscripted audio, including look-alike words (*tree* for *three*); a letter aligner places words when there is a transcript. The fine-tuning script is fixed (a 30-step check: loss 1.40 to 0.72, no NaN) but not rerun, because the tool does not need it. A full run would take 2 to 3 hours per pass on this laptop, or about an hour on a free Colab GPU.

5 Phoneme tokenisation

THE PLAN	Split speech into phoneme tokens before clustering.
MADE SENSE	PG works on segments inside words, so the tool must find the segment.
DID NOT, AND WHY	In the thesis the tokens came from text: the transcription split into words. Nothing in the audio was tokenised.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	The pronouncing dictionary says where the sound sits in each word; the phone model's alignment finds its frames in the audio.

6 IPA mapping

THE PLAN	Map words or phonemes to IPA: eSpeak in the thesis, a dictionary such as CMU in the presentation.
MADE SENSE	IPA is the shared notation, and PG tables are written in it.
DID NOT, AND WHY	A dictionary gives the citation form. The thesis's own Table 4.2 gives "there" as [ðɛr], and it would say so whether the speaker said [ð] or [d]. This step erases the variation PG ranks.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	Candidate pronunciations, written in IPA, are scored against the audio by a phone model (`wav2vec2-xlsr-53-espeak-cv-ft`), with a confidence for each token. Each sound carries a validation status (section 6).

7 CSV or VTT file

THE PLAN	Write text and IPA to a CSV, a 5×6 matrix in the framework, or a VTT subtitle file.
MADE SENSE	Plain files an analyst can open.
DID NOT, AND WHY	A VTT file needs word times. The thesis estimated them from the clip's duration; the app itself returned no word times at all. The 5×6 size was fixed by the matrix, not by the data.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	Downloads in CSV, JSON, HTML and subtitle formats, with measured start and end times for every token.

Audio clustering

8 Primary clustering

THE PLAN	K-means (and spectral clustering) on MFCC features to group similar sounds.
MADE SENSE	Unsupervised grouping could find variants without labels.

DID NOT, AND WHY	MFCCs of whole words group recordings by speaker and microphone more than by pronunciation. The number of clusters is unknown in advance, and a cluster does not say which variant it holds.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	The phone model's variant label does the grouping: every token is [θ], [t], [s] and so on, with a confidence.

9 Secondary clustering

THE PLAN	Sub-clusters by cosine similarity, at most a 5×6 matrix.
MADE SENSE	Looking for finer detail inside a variant.
DID NOT, AND WHY	The same problem as above, and the matrix size decided the clusters.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	Acoustic sub-clusters inside each variant, from the phone model's features. They barely separate anything, so the app shows them as exploratory only.

10 Matrix showing contrastiveness

THE PLAN	A confusion matrix of the sub-clusters, used for the operability analysis.
MADE SENSE	Contrast matters to PG; it is one of the six tenets.
DID NOT, AND WHY	A confusion matrix measures how much clusters overlap acoustically. PG's Contrastiveness and Disambiguity ask something else: whether a variant merges a contrast and makes words sound alike.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	The contrast matrix is kept as exploratory. Contrastiveness is marked from the Lingua Franca Core and Disambiguity from pronouncing dictionaries: [t] in *think* merges 78 common word pairs, such as *bath* and *bat*.

Standard pronunciation generation

11 Matrix processing

THE PLAN	Apply the user's ranking to the matrix. The presentation: assign a rank to each cluster by quality or relevance.
MADE SENSE	The ranking is the heart of PG.
DID NOT, AND WHY	No rule said how a candidate earns a mark on each tenet, and ranking clusters by quality is not PG's evaluation.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	Rules for each tenet that reproduce Ugorji's *think* and GOAT tableaux exactly, under strict domination. A variant competes once it has at least 3 tokens from 2 speakers. Experts can override any mark.

12 Outputs

THE PLAN	The operability table and the standard pronunciation, saved as .txt and .csv. The presentation adds a "PG operability metric" of accuracy, consistency and usability.
MADE SENSE	Two outputs, exactly as PG defines them.
DID NOT, AND WHY	Accuracy and usability measure the software, not the pronunciation, and they do not belong in PG's operability.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	Both outputs, plus warnings when a sound is not validated or the data is thin, and every token for checking by ear.

13 Operability table

THE PLAN	Ugorji's tableau: candidates in rows, tenets in columns, violation marks in the cells.
MADE SENSE	It is how PG shows its working.
DID NOT, AND WHY	Never produced.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	Produced in PG style, with an OT view. Figure 3 is the *think* table as the tool prints it.

14 Standard pronunciation

THE PLAN	The winning candidate.
MADE SENSE	PG's output component: the preferred forms earn the fewest violations.
DID NOT, AND WHY	Nothing said how sure the winner is.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	The preferred form, its tier (formal standard, allophone or outer norm), and how often it still wins when speakers are resampled and uncertain tokens redrawn.

Evaluation criteria

15 Evaluation criteria: the six tenets

THE PLAN	International acceptance, Contrastiveness, Frequency, Disambiguity, Phonetic simplicity and Pedagogic convenience, ranked I-Ac >> ContR >> FreQ >> DisaM >> Pho-S >> Ped-C.
MADE SENSE	These are Ugorji's own definitions and default ranking.
DID NOT, AND WHY	They are prose. Nothing said what evidence decides each mark, and Frequency can be read two ways.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	A source of evidence for each tenet (section 7): a knowledge base, the Lingua Franca Core, the recordings, pronouncing dictionaries and PHOIBLE. Frequency works both ways, as a setting.

Around everything

16 API, hosting and infrastructure

THE PLAN	A Node.js/Express API; AWS S3, RDS (PostgreSQL), MongoDB and a Redis queue; serving on EKS or SageMaker; ONNX; Prometheus, Grafana, CloudWatch and ELK.
MADE SENSE	An API, and containers.
DID NOT, AND WHY	It was planned before the method worked. It is far more than a team of three can run, and it costs money every month.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	One Python service serves the page and the API from one Docker image, live on Hugging Face's free hardware (section 6). One GitHub repository, deploy on push and a daily keep-awake visit are being set up.

5. The tool now

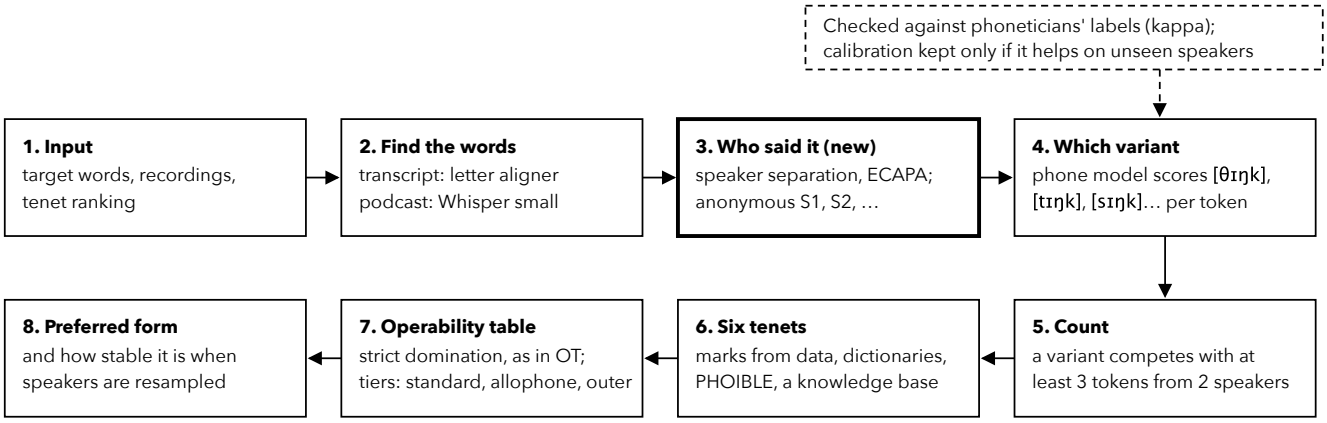


Figure 2. The full cycle as it runs now. Every step is in `backend/pg/`; the web page, the API and the command line run the same code.

The models

Job	Model	Why this one
Hear which variant was said	`facebook/wav2vec2-xlsr-53-espeak-cv-ft`	A phone recogniser trained on many languages, so it can hear [t] against [θ], not just English words. It compares candidate pronunciations of the word against the audio.
Find each word in a sentence	`facebook/wav2vec2-base-960h`	Lines a known transcript up with the audio, letter by letter.
Find words in podcasts	`openai/whisper-small`	Writes a transcript with word times; each hit is then re-aligned. It also searches look-alike words (*tree* for *three*).
Tell the speakers apart	`speechbrain/spkrec-ecapa-voxceleb`	Speaker fingerprints for short stretches of speech, grouped into speakers (section 6).

6. Results

How well it hears each sound

The only free Nigerian English recordings with pronunciations transcribed by trained phoneticians are 22 speakers in the Speech Accent Archive, all reading the same paragraph: 753 test tokens. The fairest single number is kappa, agreement beyond chance. A model that always answers the commonest variant scores 0, however good its accuracy looks.

Sound	Tokens	Accuracy	Always guess the commonest	Kappa	Status
<th> /θ/ (think)	85	82%	58%	0.68	Validated
<th> /ð/ (this), calibrated	129	81%	53%	0.64	Validated
KIT (bring, six)	112	81%	76%	0.55	Validated
NURSE (her)	86	43%	41%	0.23	Not validated
GOAT (go, snow)	68	72%	87%	0.16	Not validated
FACE (snake, train)	89	54%	72%	0.13	Not validated
FLEECE (please, meet)	183	90%	92%	0.06	Not validated

An independent reviewer was asked to break this work, not defend it. It found two fake calibrations (GOAT and FACE gave every token the same answer), a weak test ("same preferred form as the experts"), a phone model that leans towards dictionary forms (it hears Nigerian [d] as [ð] far more often than the reverse), and several bugs. All are fixed or stated plainly: calibration is now kept only when kappa rises on speakers it was not fitted on, which leaves voiced <th> alone.

What it found in Nigerian speech

On 1,519 sentences from Google's Nigerian English corpus (31 speakers):

Sound	Variant shares	Preferred (Ugorji's ranking)
<th> /θ/, 618 tokens	θ 52%, t 36%, d 5%, s 4%, f 2%. Word-initially θ 48% and stops 48%, close to ICE-Nigeria (θ 47%, t 49%).	[θ], in 100% of resamples
<th> /ð/, 925 tokens	d 57%, ð 40% once the model's bias is corrected	[ð] (99%); *d* loses on Disambiguity, not on the data
NURSE, 676 tokens (not validated)	ɜː 36%, ɛ 22%, ɔ 17%, a 15%, e 8%; the model still leans to the dictionary vowel	[ɜː]

Speaker separation

PG counts speakers, not files. The tool now cuts speech into 1.5-second windows, takes a speaker fingerprint of each with ECAPA, removes what all the fingerprints share (room, microphone, language), averages each window with its neighbours, groups them into speakers, and credits every target word to the person speaking at that moment. It was tested on made-up 4-minute panels of 2 to 5 real SLR70 Nigerian speakers, including three and four women together. The first model tried (WavLM) credited only 71% of words correctly for four women, even when told how many there were. ECAPA was tuned on 4 conversations, then tested on 8 it had never seen.

Test	Result
Held out: 8 conversations, 2 to 5 speakers	right number of speakers in 8 of 8; 100% of sampled words credited to the right speaker
Live server: a 63-second, 3-speaker conversation	3 speakers found; 5 of 5 target words credited right; 48 seconds

The test is kinder than a real podcast: each clip was recorded separately, so microphone differences help, and the speakers come from the same pool of 31.

Live deployment

The tool is live at `https://thebickersteth-voxpreference.hf.space` on Hugging Face's free CPU hardware (2 cores, 16 GB). New Docker Spaces have needed a paid plan since mid-2026; ours existed before that and keeps the free hardware, so it must not be deleted. The account has no payment method and cannot be charged. Twelve SLR70 clips from 6 speakers ran in 18 seconds (6 [θ], 4 [t], [θ] preferred). A free Space sleeps after 48 hours without visitors and wakes in a minute or two.

Podcast test

Results pending. A 43-minute Nigerian panel podcast with four hosts, two men and two women, is being analysed for every sound, once with Ugorji's ranking and once with Frequency first. The results go here when the run finishes.

7. The tenets, and Ugorji's table as the tool prints it

No PG source gives an algorithm; in the published tables the marks are the analyst's judgement. The rules below are our proposal, chosen so that both of Ugorji's evaluations come out exactly right. They are defaults the PG experts can change, not PG's own verdicts.

Tenet	A variant gets a mark when	Evidence
International acceptance	it is not an accepted international norm for the sound (for /θ/: θ, t, t̪, f are; s, z are not)	knowledge base: RP and GA, established L1 varieties, the Lingua Franca Core
Contrastiveness	it merges a contrast that matters for international intelligibility	Lingua Franca Core
Frequency	its share is under 20%, or, in "most frequent" mode, it is not the commonest variant	the recordings
Disambiguity	it makes common words sound alike (*three* and *tree*)	pronouncing dictionaries and word frequency
Phonetic simplicity	it is missing from most of Hausa, Igbo and Yoruba, or rare in the world's languages	PHOIBLE
Pedagogic convenience	it is missing from those languages and rarely used by the speakers analysed	PHOIBLE and the recordings

<th> in *think*	Share	I-Ac	ContR	FreQ	DisaM	Pho-S	Ped-C	Tier
a. tɪŋk	48.8%				*!			allophone
b. ☞ θɪŋk	47.2%					*		preferred
c. sɪŋk	4.0%	*!		*	*			outer norm
d. zɪŋk	0.0%	*!		*	*			outer norm

Figure 3. Ugorji's *think* evaluation as the tool produces it, with his default ranking: the same marks and the same winner as his slide. Shares are word-initial /θ/ in ICE-Nigeria (Oyebola, Ho & Li 2019). The reasons the tool gives: [t] makes 78 common word pairs sound alike (*bath/bat*), [θ] is absent from Hausa, Igbo and Yoruba, and [s] and [z] are outside the international norms. Shaded cells no longer matter once a fatal mark (!) is given.

What "rank by frequency" means

Most users read "Frequency first" as "the commonest pronunciation wins": if *think* came out with [θ] 49 times and that was the most, [θ] is preferred. Ugorji's *think* table only comes out right under another reading, where Frequency removes rare variants (under 20%), because in his Nigerian data [t] is slightly more common than [θ]. Both readings are in the tool. The default follows Ugorji.

8. Questions for Prof. Ugorji

- Which norms count as international, and why [t] escapes that mark in *think* while [s] does not.
- What Frequency means: a threshold or "the commonest wins"; per token or per speaker; over whose speech. This alone can flip *think* to [t].
- What separates Contrastiveness from Disambiguity, and what counts as evidence for Pedagogic convenience. The full list of 10 is in `backend/pg/RESEARCH.md`, section 6.

9. Next steps

1. Label 150 to 300 new Nigerian tokens, ideally twice, with the listening tool (`python -m pg.review`). This turns the vowels from not validated to validated.
2. Take the questions above to Prof. Ugorji before presenting automated tables as PG results.
3. Add the podcast results and check the speaker separation against the known four-person panel.
4. Try a stronger model for vowels: the Montreal Forced Aligner's English model, trained partly on Nigerian English, or formant measurements.
5. Finish the single repository with deploy on push and the daily keep-awake visit.

The method works where it has been checked. The next work is checking more of it.